MULTI-CLOUD FDE STACK · ISSUE 05 OF 08

SQS, Pub/Sub, Service Bus.

Same job, different delivery guarantees.

05

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Swipe →

What each one gives you

SERVICE	ORDERING	NOTES
SQS standard	Best effort	Visibility timeout, DLQ
SQS FIFO	Per group	Dedup window
Pub/Sub	Ordering keys	Fan-out, exactly-once option
Service Bus	Sessions	Peek-lock, dup detection

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Rules that hold everywhere

01 Idempotent consumers

A processed-message key makes replays harmless.

02 Timeout above p99

Visibility, ack deadline or lock beyond real processing time.

03 Dead-letter queue

With an alert on depth and oldest message age.

04 Small messages

Put the document in object storage and send a pointer.

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A safe document pipeline

1. Upload to object storage
2. Event to queue
3. Worker claims message
4. Process + write result
5. Ack or dead-letter

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INTERVIEW SCENARIO

“Some invoices were processed twice.”

Processing took longer than the visibility timeout, so the message came back. Raise it, and make the write idempotent.

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Queue review

- [x] Delivery semantics written down
- [x] Idempotency key on every write
- [x] Timeout set from measured p99
- [x] DLQ with alerting and redrive
- [x] Payloads in storage, not the queue

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KEY TAKEAWAY

Design for at-least-once and every cloud's queue works.

→ Make consumers idempotent

→ Set timeouts from real processing time

→ Alert on the dead-letter queue

YOUR TURN

Have you ever been bitten by a message processed twice?

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Save · Repost

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